

DEYANG (DYLAN) HSU

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EDUCATION

University of Southern California, Viterbi School of Engineering

M.S. in Computer Science | GPA: 3.88/4.0

Los Angeles, CA

Aug. 2024 – May 2026

The Ohio State University

B.S. in Electrical & Computer Engineering | GPA: 3.77/4.0 | *Magna Cum Laude*

Columbus, OH

Aug. 2018 – May 2022

TECHNICAL SKILLS

Languages Python, C, C++, Java, SQL, Bash, L^AT_EX

ML & Modeling PyTorch, TensorFlow, HuggingFace, Transformers, Mamba (SSM), GNNs/GCNs, CNNs, LoRA/PEFT, RAG, Anomaly Detection, OOD Generalization, scikit-learn, XGBoost

Systems & Data CUDA, NVIDIA A100 (AMP, `torch.compile`), SLURM, OpenMP, MPI, Docker, Kubernetes, Git, Linux, PL/SQL, Oracle, MySQL

RESEARCH & PUBLICATIONS

Research Assistant, FORTIS Lab — University of Southern California

Advisor: Professor Yue Zhao

Los Angeles, CA

Jan. 2025 – Present

- Architected a spatio-temporal wildfire prediction pipeline fusing ERA5 atmospheric reanalysis with LANDFIRE vegetation indices (LAI, NDVI) across **116,000+** time-county observations, achieving a **15% F1-score gain** over conventional baselines.
- Designed a hybrid **Mamba (SSM) + Temporal Fusion Transformer** model with **GCN** spatial message passing; engineered a custom Hybrid Poisson/BCE loss and calibrated thresholds to handle extreme class imbalance in rare fire-ignition events.
- Optimized training on **NVIDIA A100s** with mixed-precision (AMP) and `torch.compile`, delivering a **20× speedup** — cutting end-to-end training from 8+ hours to ~25 minutes and enabling rapid architecture ablation.
- Spearheaded Comparative Narrative Analysis research across **TELeR taxonomy levels 1–4**, benchmarking frontier LLMs (GPT-4, DeepSeek, Claude) on multi-narrative conflict detection; built automated evaluation pipelines scoring narrative conflict, cross-narrative overlap, and uniqueness to isolate the effect of prompt specificity on model reliability.
- Built **H-MAD**, an asynchronous multi-agent debate framework quantifying code-switching and linguistic bias in Llama 3.1 8B and Qwen 2.5 7B via novel *Linguistic Parity Score* and *Entrainment Coefficient* metrics, surfacing an LLM-as-judge limitation conflating fluency bias with factual degradation.

The Traffickers' Pitch: Detecting Deceptive Recruitment in Online Job Boards

ICWSM 2026

S. Zhou, P. Qiu, T. Salkar, L.B. Urrutia, D. Shen, **D. Hsu**, E.C. Choi, E. Ferrara — *arXiv:2605.25416*

- Co-authored a deep learning framework using contextual embeddings to isolate fraudulent recruitment patterns; engineered scalable crawling and regex pre-processing pipelines over **50K+** job postings.

PROFESSIONAL EXPERIENCE

Intel

Software Development Engineer

Chandler, AZ

June 2022 – Aug. 2023

- Root-caused and resolved P0 production defects (race conditions, memory leaks) across enterprise core microservices, increasing service uptime by **27%** and reducing average incident time-to-resolution.
- Refactored legacy manufacturing database structures with optimized **PL/SQL** and indexing strategies, substantially reducing query latency and scaling write throughput for high-volume data streams.
- Deployed automated wafer lot-tracking software adopted across global fabs, driving an **18% increase** in wafer manufacturing throughput under strict quality-compliance standards.

General Motors

Manufacturing Controls Engineer Intern

Bay City, MI

June 2021 – Aug. 2021

- Resolved hardware and software runtime faults in robotics systems and PLCs, cutting plant assembly downtime by **8%**.
- Integrated an IO-Link smart pressure sensor network via **Siemens SIMATIC STEP 7**, reducing inventory scrapping by **13%** and catching upstream calibration defects early.

PROJECTS & TEACHING

StressID: Multimodal Physiological Stress Detection

PyTorch, Wav2Vec2, MediaPipe

- Built a three-branch (audio, face, gesture) late-fusion pipeline on the **NeurIPS 2023 StressID** dataset; engineered a custom 3-block CNN feeding a 2-layer, 4-head Transformer encoder with learnable attention pooling, beating mean-pooling and frozen Wav2Vec2 baselines.
- Extracted 128-bin Mel spectrograms at 16 kHz with **SpecAugment**; validated via subject-level stratified 5-fold cross-validation with class-weighted loss under 71% class imbalance, reporting Macro-F1, AUC-ROC, MCC, and balanced accuracy.

Teaching Assistant / Grader — The Ohio State University

Advanced C++ Programming, Dept. of Electrical & Computer Engineering

Columbus, OH

Jan. 2021 – May 2022

- Graded projects for **60+** students; ran office hours debugging segfaults, pointer bugs, and memory leaks (GDB, Valgrind).